

Name: _____

Roll Number: _____

Quiz-1

Max. Time: 15 min

Max. Points: 20

Note: Solve all parts. Limit your written responses to the provided space.

Q.1. [8] Choose by putting a check mark on the most appropriate option. Note: No cutting/overwriting is allowed.

i. In some cases, a matrix may be row reduced to more than one matrices in reduced echelon form by using different sequences of row operations.

(A) True (B) False

ii. Existence of a free variable implies consistency of a linear system

(A) True (B) False

iii. The linear system $\mathbf{Ax} = \mathbf{b}$ is consistent only if the augmented matrix $[\mathbf{A} \ \mathbf{b}]$ has a pivot position in every row.

(A) True (B) False

iv. The echelon form of a matrix is unique.

(A) True (B) False

v. The pivot positions in a matrix depend on whether row interchanges are used in the row reduction process.

(A) True (B) False

vi. A 3×5 coefficient matrix of a linear system with pivots in every row can be inconsistent.

(A) True (B) False

vii. An overdetermined linear system (more equations than the unknowns) is always inconsistent.

(A) True (B) False

viii. The points corresponding to $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ -7 \end{bmatrix}$ lie on a line through the origin.

(A) True (B) False

Q.2. [7+5]

a) Row-reduce the matrix $\mathbf{A}$ to the reduced echelon form.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 5 & -6 & -2 & 7 \\ -5 & 7 & 8 & 0 \end{bmatrix} \sim \dots \sim \begin{pmatrix} 1 & 0 & 0 & \frac{395}{109} \\ 0 & 1 & 0 & \frac{169}{109} \\ 0 & 0 & 1 & \frac{99}{109} \end{pmatrix}$$

Grading notes:

1. Give 4 marks to those who attempted it, tried various row operations, and exposed two pivot positions
2. Give 6 marks if '1' is true and three pivot positions were revealed
3. Give 7 marks if '1' and '2' and proper format of reduced echelon form even with decimal numbers

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b) Using the matrix $\mathbf{A}$ from part 'a', check if the linear system $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{b} = \begin{bmatrix} 10 \\ 3 \\ 0 \end{bmatrix}$, is consistent.

$$[\mathbf{A}|\mathbf{b}] = \begin{pmatrix} 1 & 2 & -3 & 4 & 10 \\ 5 & -6 & -2 & 7 & 3 \\ -5 & 7 & 8 & 0 & 0 \end{pmatrix} \sim \dots \sim \begin{pmatrix} 1 & 2 & -3 & 4 & 10 \\ 0 & -16 & 13 & -13 & -47 \\ 0 & 0 & \frac{109}{16} & \frac{99}{16} & \frac{1}{16} \end{pmatrix}$$

The system is consistent as there is no pivot position in the augmented column.

Grading notes:

1. Give 3 marks if augmented matrix is correct
2. Give zero marks if previously reduced echelon form from part 'a' is used to augment with the vector $\mathbf{b}$
3. Give 4 marks if '1' and proper steps to get to row echelon form
4. Give 5 marks if '1', '3', and correct result as shown above (even in decimal form)

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